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SHORT-PAPER

Robot Housekeeper Cart: Empowering Housekeeping Work through Etiquette-Based Interactions in Hotels

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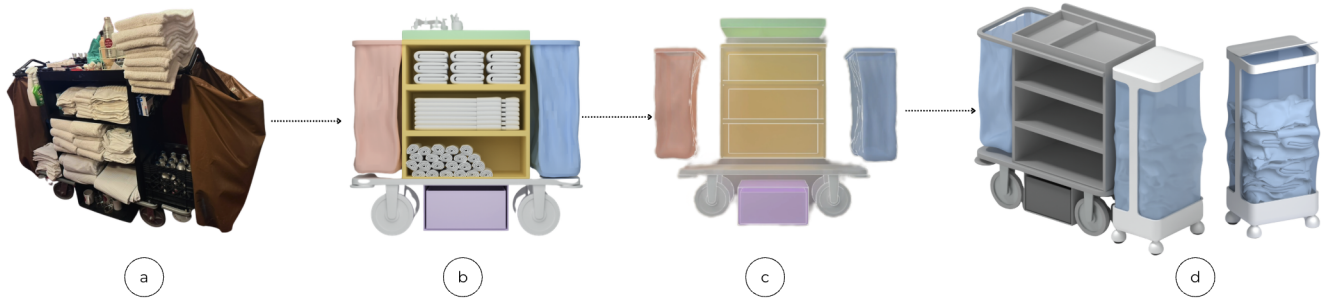


Figure 1: Concept modeling of the modular housekeeper cart. (a) a traditional housekeeper cart; (b) organization division of the cart; (c) division and modularization of each unit; (d) example of a laundry module.

Abstract

Housekeeper carts are essential in hotel operations, supporting the maintenance of the hotel's physical environment and services. While housekeeping staff are their main users, carts are also highly visible to guests, making them not only tools but also sites where hotel experiences are shaped. This project re-designs housekeeper carts to address both their functional and experiential value to primary users and bystanders. We present a modularized cart with an in-depth development of the laundry module. Considering hotels' need for trustworthy and polite interactions, we designed non-verbal behaviors that allow the robot to express etiquette.

CCS Concepts

• Human-centered computing → Collaborative interaction.

Keywords

Human-Robot-Interaction, Hotel Robot, Non-verbal Interactions

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1 Introduction

Housekeeper carts are central to the everyday operations of hotels, supporting the upkeep of rooms, linens, and guest environments. Yet despite their ubiquity, these carts often create friction for both workers and guests. Our observational study revealed that conventional carts are heavy, difficult to maneuver, and prone to causing strain on housekeeping staff and damage to hotel walls and corridors. Previous studies have shown that hotel housekeepers face substantial ergonomic and health challenges, including musculoskeletal strain and repetitive-movement injuries, resulting from daily cleaning tasks [1, 4]. At the same time, these carts occupy highly visible public spaces, becoming part of the guest experience, whether intentionally or not. This dual identity of the housekeeper cart as both back-of-house equipment and front-facing experiential touchpoints raises design challenges seldom addressed in existing hotel infrastructure. We therefore reimagine housekeeper carts not only as functional tools but also as interactive agents capable of participating in the social texture of hotels.

In this project, we redesign the housekeeper cart as a modular robotic system and develop a laundry robot module that integrates expressive, non-verbal interaction capabilities [7]. Previous studies have shown that humans can reliably interpret signals from non-humanoid robots based solely on motion patterns, suggesting that even subtle movements can effectively convey intent without an anthropomorphic appearance [11]. Drawing on observations of housekeeper workflow and interviews with hotel managers, we translated key etiquette practices including proactive greetings, anticipating needs, clear communication, and showing gratitude into non-verbal robot behaviors [3]. We then explored these behaviors through sketching, 3D modeling, animation, and prototyping,

focusing particularly on lid movement as a site for expressive communication.

We further evaluated the robot's behavior through iterative testing in controlled and real-world environments. Early findings show that people consistently perceive the robot as *polite*, yet often express curiosity and uncertainty about its intentions and operational boundaries [2]. These insights informed refinements to movement, speed modulation, and physical signaling. A planned deployment in a functioning hotel will extend these explorations into the robot's intended context.

By combining functional redesign with etiquette-informed non-verbal HRI, this work contributes: a design framework for translating hotel etiquette into non-verbal robot behaviors; a modular cart-robot design tailored to housekeeping workflows; and empirical insights from iterative prototyping and in-the-wild testing.

2 Study in the Hotel

We carried out observational studies and expert interviews to better understand the housekeeping workflow and hotel etiquette training to inform our design.

In shadowing and observational studies of the housekeeping team, including the housekeepers, housemen, and managers, we discovered a need to redesign the housekeeper cart. We found that the current housekeeper carts are always messy, bulky, and heavy for the housekeepers to push around (Fig. 1a), affecting both the workers' physical health [4], causing damage to the hotel environment, and indirectly impacting guest experiences. For instance, we found that the workers always need to back up the carts and use momentum to push the cart again, largely leaning their body weight on the carts to generate enough power to move forward. Because of its gigantic size, the cart often hits hallway walls, damaging the walls or leaving dirty marks.

Housekeeper carts are not only an internal tool but also an integral part of the hotel's externally facing operations. Through conversations with hotel managers, we learned about the following etiquette goals, their training for employees, and how these inform our design of the robot behavior [9]. Among the many hotel etiquettes, there are four aspects that are most essential to our study: proactive greetings, anticipating needs, clear and proper communication, and showing gratitude (Table 1). In hotel training



Figure 2: Selected lid opening and movement designs.

for their employees, these etiquettes often manifest in the employees' language and behavior [5]. In hallways on guestroom floors, hotel management is cautious about the addition of sound and light. Therefore, we grounded our translation and design of the robot etiquette in non-verbal and light based interactions.

3 Design Practice

From our study in the hotel, we recognized that the core users of the cart are the housekeepers, with hotel guests and other hotel employees being bystanders. The housekeeper carts operate in the hallways and laundry rooms. From the observation of the sectional organization of the original laundry cart, we envision a modular cart (Fig. 1b) in which each component can operate independently when needed (Fig. 1c). To explore what such a modular design means in a hotel setting, we started with the laundry module (Fig. 1d) to develop more interaction design in greater depth.

To explore our design, we followed a process of brainstorming, sketching, 3-D modeling, and animation to visualize and test our ideas before physical prototyping.

The lid of the laundry bot became a site for non-verbal interactions. We explored different combinations of lid shapes, openings, and movements. For instance, (Fig. 2a) showcased a 1-piece lid that opens to signal noticing, (Fig. 2b) a 2-piece horizontal-cut lid waving, and (Fig. 2c) a 2-piece short vertical-cut lid to express sadness. These different shapes and cuts of the lid enable various possibilities in movements to express etiquette.

Taking into account the etiquette insights and mapping them to our design, we developed an animated storyboard (Fig. 3) to showcase the non-verbal behavioral designs of the robot and highlight the details. When considering proactive greetings, we also refer

Table 1: Mapping Hotel Etiquette to HRI Design Strategies

Etiquette	Employee Practices	HRI Design on Laundry Bot
Proactive Greetings	always be the first to acknowledge a guest with a warm and inviting demeanor.	signaling acknowledgment without interrupting.
Anticipating Needs	identify and fulfill guest needs before they ask.	the bot anticipates housekeeping workflow and repositions itself near staff before towels or transport are needed
Clear & Proper Communication	using simple language, listening attentively, and providing accurate information.	the bot uses explicit speed modulation and motion cues to make its intended motion clear
Show Gratitude	closing interactions with thanks or appreciation	the bot ends hand-offs with a soft lid movement pattern to signal "Goodbye"

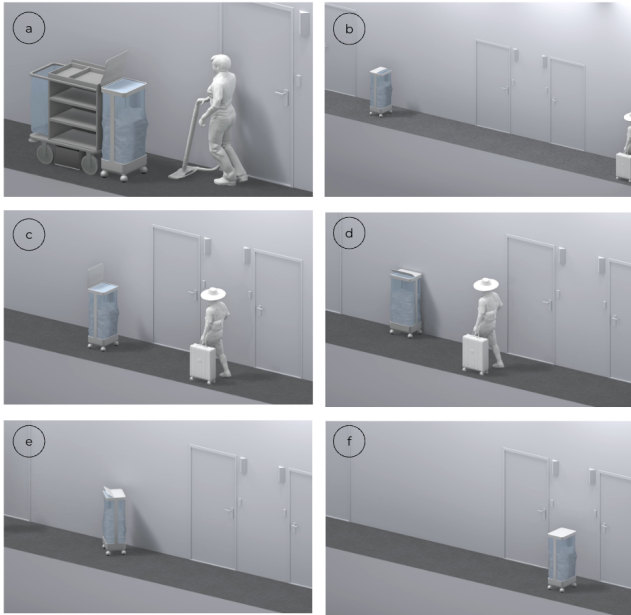


Figure 3: Interaction design: (a) The bot lifts its lid to signal “Goodbye”. (b) It recognizes an approaching guest and signals noticing. (c) The lid opens wider to gesture “Hi.” (d) The bot backs up and to give more space. (e) After passing, the bot checks on the guests. (f) The bot then continues its route.

to the “ten and five rule” learned from an interview with a hotel manager: “staff coming within 10 feet of a guest should smile and make eye contact, while those coming within five feet should add a warm verbal greeting.” In our design, the bot will notice a guest at a distance with a subtle opening of its lid (Fig. 3b). When the guest moves toward the robot, it will initiate a greeting gesture, “hi” (Fig. 3c). To provide clear and proper communication, the bot will also adjust its speed when getting closer to the guest. When passing by, the bot will slowly and politely move close to the wall to provide more space for the guest to walk by (Fig. 3d). To show gratitude and a proper “goodbye”, the bot’s body angle will follow the guest’s leaving direction and use a slight lift to showcase the checking status (Fig. 3e). The more fluid interactions can be found in our video complementary to this paper.

4 Prototyping

With our research questions and interaction design requirements in mind, we selected the *WOWLIVE 100 L Rolling Laundry Hamper* as the prototyping platform. This choice was guided by several practical criteria: a pre-existing wheeled base, a robust metal frame paired with a soft internal laundry bag, and a two-piece modular construction (frame and bag) that allowed for flexible modification during development.

The hamper has a capacity of 100 L and weighs 2.9 kg. Although the manufacturer does not specify a maximum load rating, comparable 100 L hampers typically support up to 25 kg of laundry. These size and weight considerations informed our requirement for



Figure 4: Laundry Module Prototype

a drive system capable of delivering sufficient torque for both basic mobility and realistic operational conditions.

To meet these mobility requirements, we adopted the hoverboard-ODrive conversion methodology developed by Wendy Ju’s Interaction Research Lab. Prior work by Ju et al. demonstrates the feasibility of repurposing consumer hoverboards into low-cost, high-performance mobile robot platforms [8]. Specifically, we selected the Replay Aeroglide Hoverboard and mounted its wheels onto the laundry cart frame. We designed and 3D-printed custom brackets aligned with the wheel assembly’s existing mounting holes and secured the assembly to the cart frame using zip ties (Fig. 4). We then followed Ju et al.’s documented procedures to establish the control system connecting the hoverboard motors, ODrive motor controller, Raspberry Pi microcontroller, and laptop interface.

5 Implementation In the Wild

We conducted a multi-phase user testing protocol consisting of two Wizard of Oz (WoZ) studies [10], one in the studio and the other in the wild [6].

For WoZ in the studio, we conducted initial testing in the hallway adjacent to our studio space. We recruited a sample of three participants (P1-P3) for structured interaction sessions. Participants received a briefing on the robot’s design intent and operational scenario, and were then asked to role-play as hotel guests navigating the corridor.

Across participants, the robot was consistently perceived as polite. P1 and P2 both recognized the cart as service-oriented and expressed curiosity about its contents without attempting to open the lid. While P2 successfully interpreted the robot’s movement intentions, they were unsure whether touching it was permissible. P3 also described the robot as polite but was confused by the hoverboard’s wheel lights, which led them to attempt to move the cart manually. They further noted that backward movement communicated yielding intentions more clearly than brief stopping. Overall, participants’ curiosity about the contents co-occurred with issues of action clarity and misleading wheel lighting. In response, we concealed the hoverboard lights and implemented a two-tier speed control system that enables more legible, gradual yielding behavior.

With these learnings, we deployed WoZ in the wild in an uncontrolled public environment to observe spontaneous reactions without prior context or instruction. The deployment site (Fig. 5) was

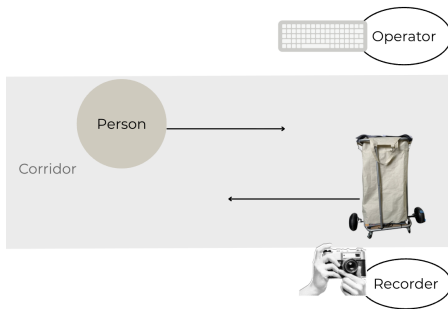


Figure 5: Field Testing Set Up

a corridor in a university building. The robot operator positioned themselves in the alcove, while a video recorder was stationed in the study area; both appeared to be engaged in academic work.

When pedestrians approached, the robot initiated movement towards them. Upon reaching proximity, the robot performed a yielding action and then resumed travel after the pedestrian passed. The robot then returned to its starting position. Due to spatial constraints, the robot only interacted with traffic from one direction.

In this study, seven individuals passed through during the observation period; five successful interactions occurred (the robot yielded before individuals passed with no interaction). People often did not notice the robot until it moved. We observed a range of reactions, including surprise, curiosity, indifference, and hesitation among these pedestrians (Fig. 6).

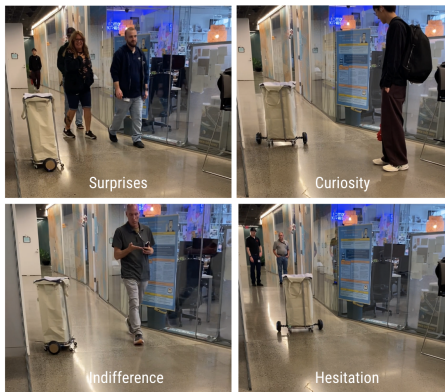


Figure 6: A selection of reactions towards the robot.

6 Discussion and Future Works

This work introduced a modular robotic housekeeper cart designed to support hotel staff through etiquette-informed, non-verbal interaction strategies. Our iterative prototyping and Wizard-of-Oz studies demonstrated that a robot with limited degrees of freedom can communicate high-dimensional social intent, such as greeting or yielding, solely through the nuanced control of moving speed and in hotel contexts.

Looking ahead, we aim to deploy the laundry module in real hotel environments and study how the robot integrates into the day-to-day rhythms of housekeeping work. This next phase includes close collaboration with housekeepers as primary users and with managers as planners and coordinators who set cleaning goals, routes, and priorities. Future work will also examine how the system performs across different hotel types and working conditions, recognizing that hospitality practices vary widely across budget, boutique, and luxury settings. These investigations will help us understand how interaction load, workflow structures, and spatial constraints shape the design of housekeeping robots. Finally, we plan to extend the interaction repertoire beyond greetings and yielding behaviors, developing additional modes of communication that align with the norms and sensitivities of hotel environments. Together, these efforts will deepen our understanding of how service robots can meaningfully empower hospitality workers while maintaining the guest experience central to hotel operations.

Acknowledgments

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